

TECHNICAL SUMMARY SHEET: WAREHOUSE LOGISTICS AGENT

Course: CS-AI-2026	Group ID: Group #05 (Team Antigravity)	Track: Track 1: Warehouse Logistics Agent
Members: Edrin Mathew & Team	Repository: github.com/edrinmathew1/logistics-agent	Algo: A* Search with Weighted Terrain

1. PEAS Framework Matrix

Component	Definition & Operational Details
Performance Measure	Minimized terrain-weighted path cost, search efficiency (nodes expanded), battery constraints (refill detours), dynamic obstacle replanning success rate.
Environment	12x12 Grid Warehouse, fully observable, static & dynamic shelf obstacles, weighted terrain (Narrow Aisles cost 3.0 vs Normal Floor cost 1.0), discrete & deterministic.
Actuators	4-Directional Grid Transitions (Up, Down, Left, Right); Package Pick-up action; Battery Recharge action at Charging Station (1, 4).
Sensors	Full state perception (Forklift position, battery level, shelf obstacle map, package coordinates P1/P2, loading bay location).

2. Core Algorithmic Formulation

State Space S: $S = (x, y, \text{battery_level})$ in $\{0..11\} \times \{0..11\} \times \{0..25\}$
Initial State S0: $S_0 = (1, 1, 25)$ with empty cargo hold at Forklift Start
Goal Test: Sequential multi-leg check: Leg 1 == P1 (10, 2) -> Leg 2 == P2 (2, 10) -> Leg 3 == Bay (10, 10)
Path Cost g(n): Accumulated terrain-weighted step cost: Normal Floor = 1.0, Narrow Aisle = 3.0
Heuristic Equations h(n):

- Manhattan Distance (Primary):** $h_{\text{manhattan}}(n) = |x_1 - x_2| + |y_1 - y_2|$ [Strictly Admissible & Consistent: min step cost ≥ 1.0]
- Euclidean Distance (Variant):** $h_{\text{euclidean}}(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- Dijkstra Benchmark:** $h_{\text{dijkstra}}(n) = 0.0$

Dynamic Replanning & Battery Logic: If path blocked at runtime, re-run A* live from S_{curr} . If battery < required cost, route through Charging Station (1, 4) to refill to 25.

3. Complexity Analysis & Empirical Metrics

Theoretical Complexity: Time $O(b^d)$ worst-case / $O(|E| \log |V|)$ using Min-Heap priority queue; Space $O(b^d)$ for Open Priority Queue & Closed Set storage.

Heuristic Variant	Weighted Path Cost	Nodes Expanded	Execution Time	Search Efficiency Gain
Manhattan A*	52.0 units	120 nodes	0.82 ms	29.8% fewer nodes vs Dijkstra
Euclidean A*	52.0 units	136 nodes	0.95 ms	20.4% fewer nodes vs Dijkstra
Dijkstra (h = 0)	52.0 units	171 nodes	1.28 ms	Baseline (Unguided Search)